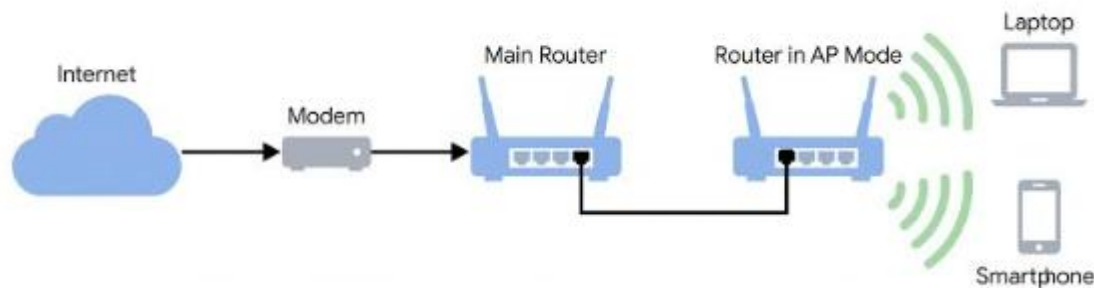


1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

Network Component Connections :



A standard network follows a hierarchical flow from the internet service provider down to your personal devices:

- **Internet / ISP:** The external fiber, cable, or DSL line coming into your house.
- **Modem:** Connects directly to the ISP line via coax or fiber. It translates the incoming signal into a standard digital internet connection.
- **Router:** Connects to the modem's WAN port via an **Ethernet cable**. It acts as the network director, assigning local IP addresses and managing traffic.
- **Network Switch:** Connects to one of the router's LAN ports via **Ethernet**. It expands the number of available physical Ethernet ports on your network.
- **Wireless Access Point (AP):** Connects to the switch via **Ethernet** to extend or broadcast a strong Wi-Fi signal to areas the main router can't reach.
- **Smart Devices:**
 - **Laptop:** Connected wirelessly to the **Access Point** or main router via **Wi-Fi**.
 - **Smartphone:** Connected wirelessly to the network via **Wi-Fi**.

2 List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

- **Modem**
 - **Main Function:** Brings the internet into your home by translating the signal from your Internet Service Provider (ISP) into a digital format your local devices can understand.

- **Real-Life Example:** Tucked away in a **utility closet or next to the cable jack** in your living room, acting as the starting point for your home internet connection.
- **Router**
 - **Main Function:** **Directs traffic and shares the internet** between your modem and all your internal home devices, assigning a unique local address (IP address) to each one.
 - **Real-Life Example:** Placed **centrally in a living room** to distribute an internet connection to everyone in the family simultaneously.
- **Switch**
 - **Main Function:** **Adds more plug-in ports** to your network, allowing multiple wired devices to talk directly to each other and share a single connection to the router.
 - **Real-Life Example:** Sitting **behind a home theater setup or gaming console desk** to plug in a PlayStation, a Smart TV, and an audio receiver all at once using cables.
- **Access Point (AP)**
 - **Main Function:** **Extends wireless coverage** by broadcasting a Wi-Fi signal from a wired connection, helping eliminate dead zones in large spaces.
 - **Real-Life Example:** Mounted on the **ceiling of a long hallway or upstairs office** to ensure you get full Wi-Fi bars even when far away from the main router.
- **Network Cable (Ethernet)**
 - **Main Function:** **Carries data physically** between devices at high speeds with maximum stability and zero wireless interference.
 - **Real-Life Example:** Running **from your desktop computer directly into the wall jack** to guarantee a lag-free connection during video calls or online gaming

3 .Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

1. Ethernet Cable



Common Usage: Commonly used to connect devices like computers, routers, and switches within local area networks (LANs) for stable high-speed wired internet connections.

2. Coaxial Cable



- **Common Usage:** Commonly used to transmit television signals from antennas or satellite dishes directly into TVs and set-top boxes, as well as for broadband cable internet.

3. Fiber Optic Cable



- **Common Usage:** Commonly used for high-bandwidth, long-distance communication links, such as underground or undersea telecommunication grids and ultra-fast Fiber-to-the-Home (FTTH) internet connections.

4 Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Feature	Router	Switch
Main Use	Connects multiple distinct networks together (e.g., joining your home network to the internet)	Connects multiple devices within a single, local network to share resources
Typical Location	At the edge of a network (acting as the gateway between networks)	Inside a local area network (LAN) (connecting computers, printers, and servers)

Data Handling	Directs data packets based on IP addresses across different networks	Directs data frames based on MAC addresses to specific devices in the same network
OSI Model Layer	Operates at Layer 3 (Network Layer)	Operates at Layer 2 (Data Link Layer)